

Bitmain Antminer Diagnostic Checklist

Technician Name: _____ Date: _____
Miner Model: _____ Serial Number: _____
Firmware Version: _____ Reported Issue: _____

1. External Inspection

- ☐ Check for physical damage to case/chassis
- ☐ Inspect power cables for damage or fraying
- ☐ Check power supply unit for physical damage
- ☐ Inspect network cable and connection
- ☐ Look for signs of overheating (discoloration, melting)
- ☐ Check for unusual odors (burnt electronics)
- ☐ Inspect fans for dust accumulation or damage
- ☐ Verify all screws and fasteners are present
- ☐ Inspect external connectors for corrosion or damage
- ☐ Check for any liquid damage or residue

2. Initial Power Test (Unplugged)

- ☐ Verify power button functionality (if applicable)
- ☐ Check power supply output connectors
- ☐ Measure PSU output voltages with multimeter
 - ☐ 12V: _____ V (Target: 11.8-12.2V)
 - ☐ 5V: _____ V (Target: 4.8-5.2V)
 - ☐ 3.3V: _____ V (If applicable) (Target: 3.2-3.4V)
- ☐ Check PSU fan functionality
- ☐ Inspect PSU internal components if suspect

3. Internal Hardware Inspection

- ☐ Remove covers following proper ESD procedures
- ☐ Inspect control board for:
 - ☐ Burnt or damaged components
 - ☐ Bulging or leaking capacitors
 - ☐ Corrosion or residue
 - ☐ Cold solder joints
 - ☐ Broken traces

- ☐ Inspect all hash boards for:
 - ☐ Burnt ASIC chips
 - ☐ Damaged components
 - ☐ Corrosion on connectors
 - ☐ Physical damage to board
 - ☐ Burned traces or areas
- ☐ Check all internal connections and cables
- ☐ Inspect internal power distribution
- ☐ Verify heat sink and thermal pad condition
- ☐ Check fan connectors and wiring

4. Powered-On Diagnostics (Safe Mode)

- ☐ Connect power (not to mining pool)
- ☐ Observe startup sequence:
 - ☐ Fan startup and operation
 - ☐ LED indicator patterns
 - ☐ Listen for unusual sounds
- ☐ Record fan behavior:
 - ☐ All fans operating: Yes/No
 - ☐ Fans at proper speed: Yes/No
 - ☐ Unusual noise: Yes/No
- ☐ Observe control board LED indicators:
 - ☐ Power LED: On/Off/Flickering
 - ☐ Status LED: On/Off/Flickering
 - ☐ Network LED: On/Off/Flickering
 - ☐ Hash board status LEDs: Notes: _____

5. Network Connectivity Check

- ☐ Connect the Ethernet cable
- ☐ Verify the link and activity LEDs are active
- ☐ Scan the network to find miner's IP
- ☐ Ping miner IP address: Success/Fail
- ☐ Access web interface: Success/Fail
- ☐ Login to the web interface: Success/Fail
- ☐ Record IP address: _____

6. Web Interface Diagnostics

- ☐ Record temperature readings:
 - ☐ Control board: _____ °C
 - ☐ Hash board 1: _____ °C
 - ☐ Hash board 2: _____ °C
 - ☐ Hash board 3: _____ °C
- ☐ Check the hash board detection:
 - ☐ Number of boards detected: _____/3
 - ☐ Chips detected per board: /
- ☐ Check the Kernel log for errors
- ☐ Record system uptime: _____
- ☐ Verify current firmware version: _____
- ☐ Check for error messages: _____

7. Hash Board Diagnostics

For each hash board (repeat for all boards, typically 3):

Hash Board 1

- ☐ Visual inspection complete
- ☐ Check voltage at test points:
 - ☐ Core voltage: _____ V (Target: _____ V)
 - ☐ IO voltage: _____ V (Target: _____ V)
- ☐ Number of working chips detected: /
- ☐ Hash rate reported: _____ TH/s
- ☐ Stable operation for 5+ minutes: Yes/No
- ☐ Temperature within specs (usually <85°C): Yes/No
- ☐ Board-specific issues: _____

Hash Board 2

- ☐ Visual inspection complete
- ☐ Check voltage at test points:
 - ☐ Core voltage: _____ V (Target: _____ V)
 - ☐ IO voltage: _____ V (Target: _____ V)
- ☐ Number of working chips detected: /
- ☐ Hash rate reported: _____ TH/s
- ☐ Stable operation for 5+ minutes: Yes/No
- ☐ Temperature within specs (usually <85°C): Yes/No
- ☐ Board-specific issues: _____

Hash Board 3

- ☐ Visual inspection complete
- ☐ Check voltage at test points:
 - ☐ Core voltage: _____ V (Target: _____ V)
 - ☐ IO voltage: _____ V (Target: _____ V)
- ☐ Number of working chips detected: /
- ☐ Hash rate reported: _____ TH/s
- ☐ Stable operation for 5+ minutes: Yes/No
- ☐ Temperature within specs (usually <85°C): Yes/No
- ☐ Board-specific issues: _____

8. Performance Testing

- ☐ Connect to test mining pool
- ☐ Run for minimum 30 minutes
- ☐ Record stable hash rate: _____ TH/s
- ☐ Record expected hash rate: _____ TH/s
- ☐ Performance percentage: _____ % (Target: >95%)
- ☐ Record reject rate: _____ % (Target: <1%)
- ☐ Record power consumption: _____ W
- ☐ Calculate efficiency: _____ J/TH
- ☐ Thermal stability: Pass/Fail
- ☐ Fan stability during full load: Pass/Fail

9. Firmware Assessment

- ☐ Current firmware appropriate for model: Yes/No
- ☐ Firmware update required: Yes/No
- ☐ Recovery mode functioning: Yes/No
- ☐ Configuration settings appropriate: Yes/No
- ☐ Security settings configured properly: Yes/No

10. Problem Diagnosis Summary

- ☐ Primary issue identified: _____
- ☐ Secondary issues identified: _____
- ☐ Root cause assessment: _____
- ☐ Components requiring repair/replacement:
 - ☐ _____
 - ☐ _____

11. Repair Plan

☐ Recommended actions:

- ☐ _____
- ☐ _____
- ☐ _____

☐ Estimated repair time: _____

☐ Parts required: _____

☐ Estimated cost: _____

☐ Repair priority: High / Medium / Low

12. Additional Notes

- ☐ _____
- _____
- _____
- _____

13. Final Recommendations

- ☐ Proceed with repairs
- ☐ Beyond economic repair - recommend replacement
- ☐ Partial repair recommended
- ☐ Additional diagnosis required
- ☐ Return to manufacturer (RMA)
- ☐ No issues found - return to service

Diagnostic Completed By: _____

Signature: _____

Date/Time Completed: _____

Repair Approved By: _____

Date: _____